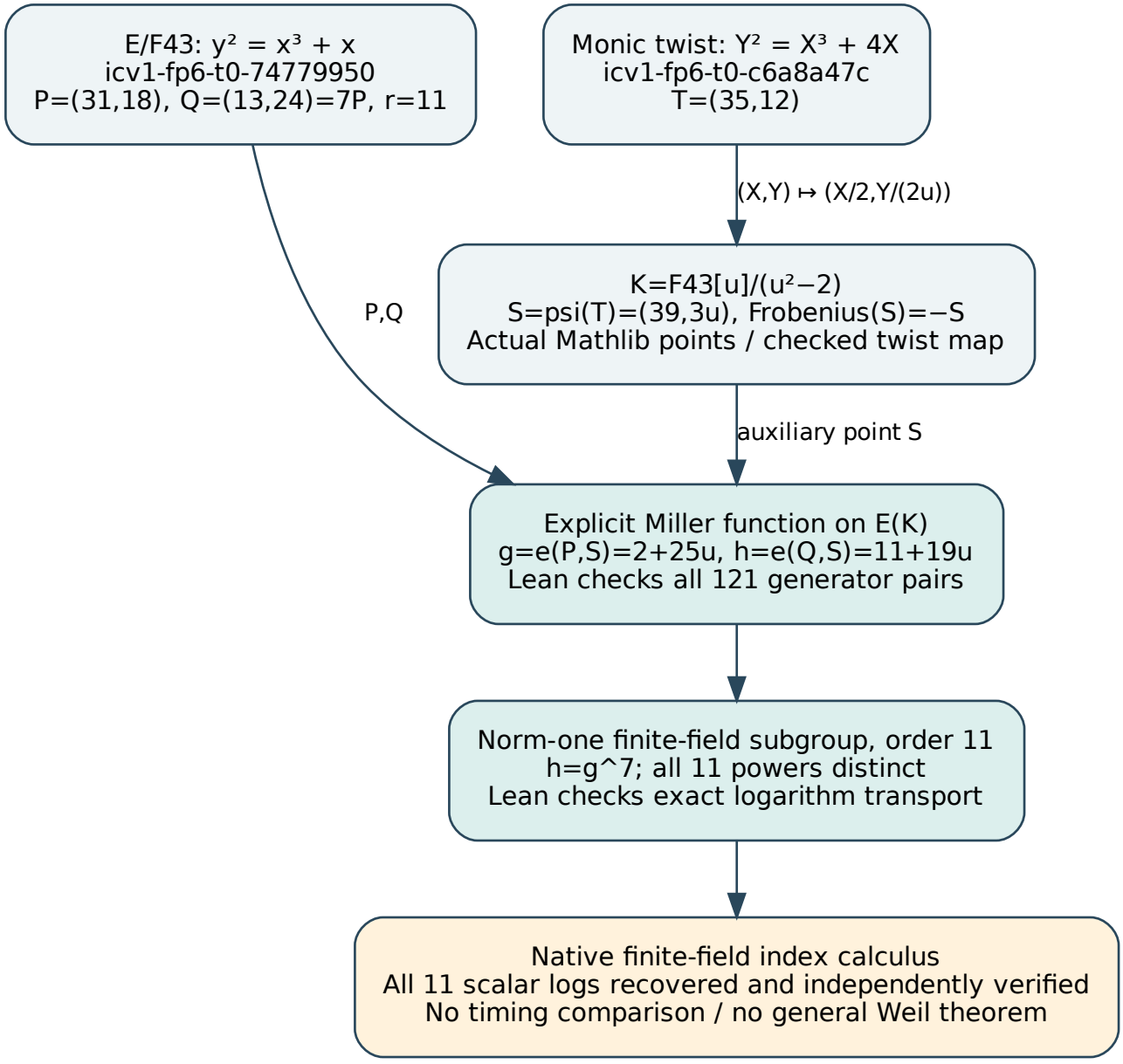
E/F43: y² = x³ + x
icv1-fp6-t0-74779950
P=(31,18), Q=(13,24)=7P, r=11
Monic twist: Y² = X³ + 4X
icv1-fp6-t0-c6a8a47c
T=(35,12)
(X,Y) ↦ (X/2,Y/(2u))
K=F43[u]/(u²−2)
S=psi(T)=(39,3u), Frobenius(S)=−S
Actual Mathlib points / checked twist map
P,Q
auxiliary point S
Explicit Miller function on E(K)
g=e(P,S)=2+25u, h=e(Q,S)=11+19u
Lean checks all 121 generator pairs
Norm-one finite-field subgroup, order 11
h=g^7; all 11 powers distinct
Lean checks exact logarithm transport
Native finite-field index calculus
All 11 scalar logs recovered and independently verified
No timing comparison / no general Weil theorem